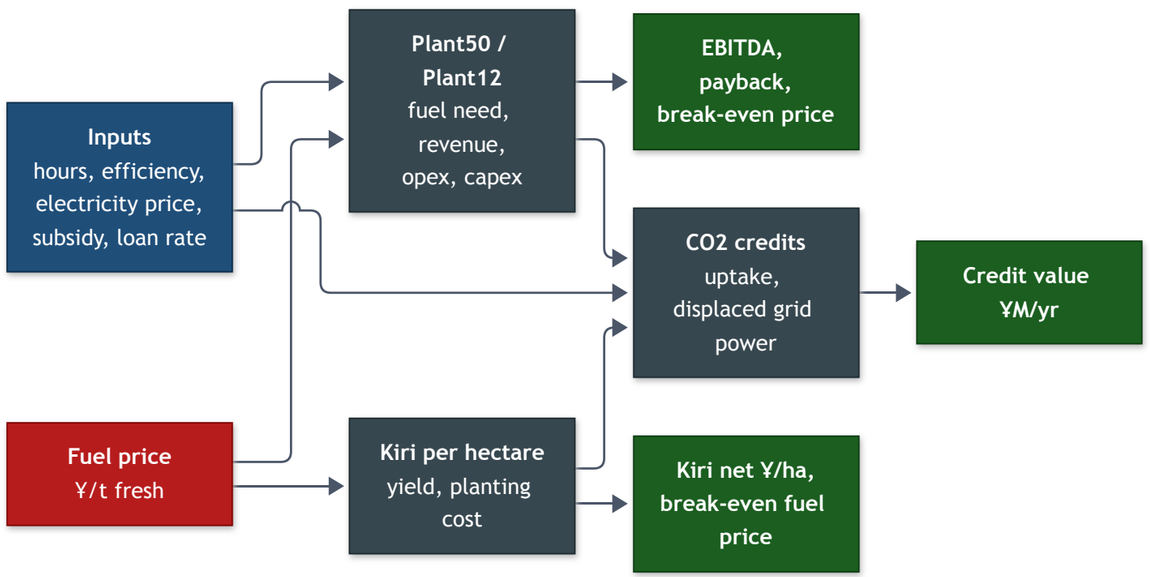
Kiri vs Sugi CHP Financial Model: How the Calculations Fit Together

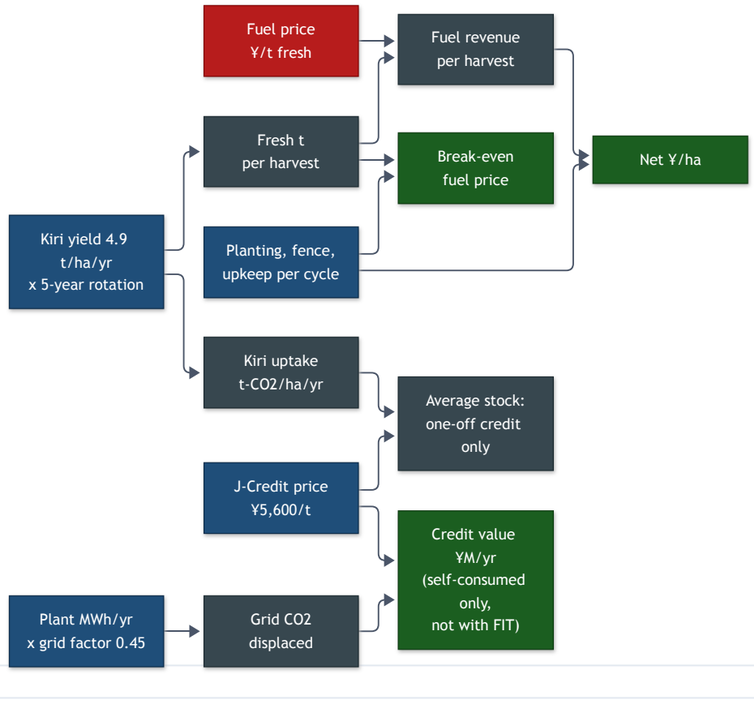
Working model v1.0 (2026-10-31). Blue = inputs, grey = calculations, green = results, red = decisive lever. All costs are placeholders, not quotes.

v1.0 · 2026-10-31

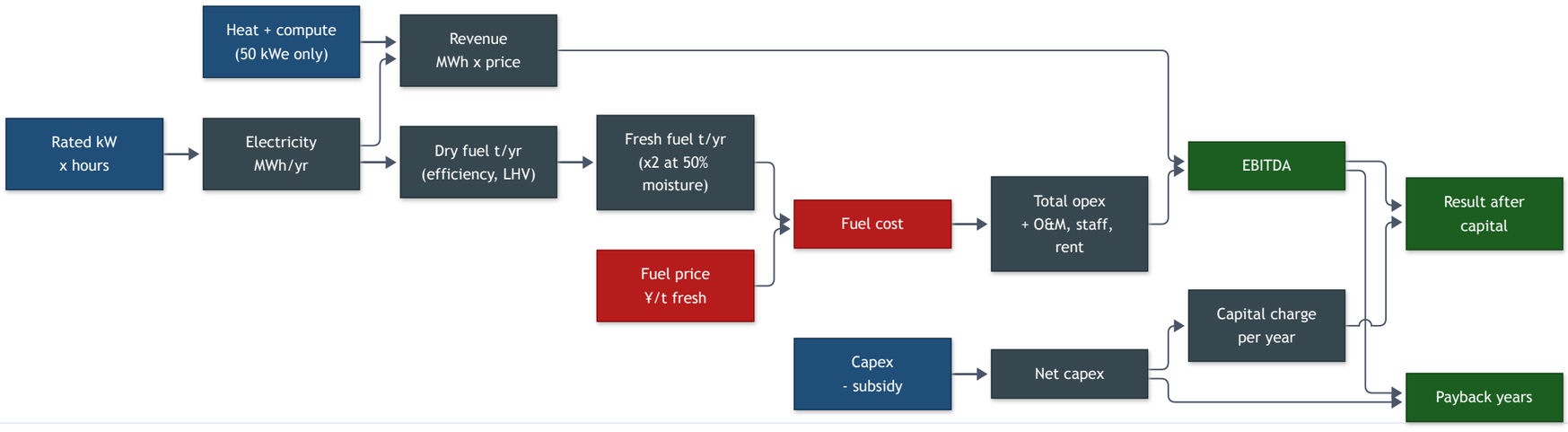
1. Overview: what feeds what



2. Kiri per hectare and CO2 credits



3. Plant sheets (50 kWe and 1.2 MWe)



Key results (placeholders, base case)

- 50 kWe, electricity only: EBITDA about -¥8M/yr (break-even about ¥53/kWh before capital)
- 50 kWe with heat + compute: payback about 5 yr (2/3 subsidy) to 15 yr (none)
- 1.2 MWe: positive only if fuel ≤ about ¥5,000/t fresh
- Kiri as fuel: about ¥0.4M revenue per ha per cycle vs about ¥2M to plant
- CO2: 50 kWe displaces about 180 t/yr, about ¥1M/yr at ¥5,600/t

Calculations: Excel model kiri_sugi_chp_financial_model_2026-10-31.xlsx (same folder; sheets Inputs, Plant50, Plant12, Kiri_ha, CO2, Notes). Change the blue cells to test other fuel prices, subsidies or yields.